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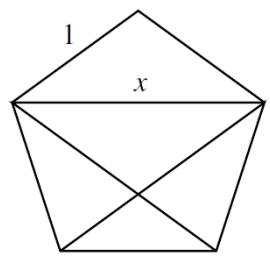
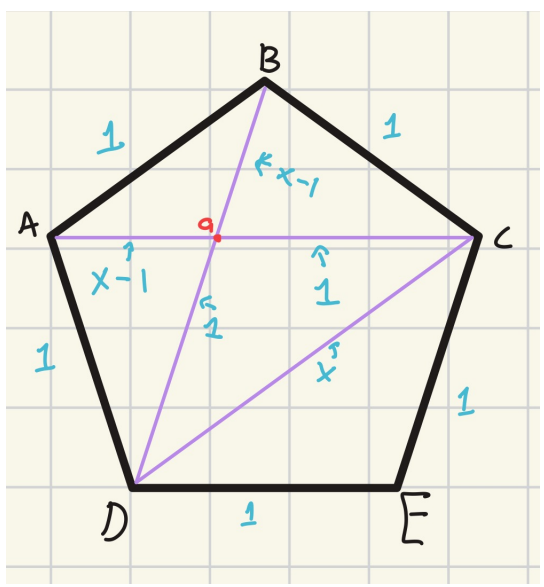
MATH 6400: Quiz #3
Due: January 28, 2026 @ 23:59

1. Assume the figure below is a regular pentagon with each side being length 1. Prove that x is the *Golden Ratio*; that is $x = \frac{1+\sqrt{5}}{2}$. Hint: Find similar triangles and parallel lines to show that x satisfies

$$\frac{x}{1} = \frac{1}{x-1},$$



Solution: We can see since all sides of the pentagon have length 1, this means any line between two non-adjacent corners will have the same length, x. We can then label the points as outlined in the figure below



From here, we can construct diagonals AC and BD, and denote the intersection of these lines a. By transferring the line DE to the line created by aC as described in Euclid I.2, we can see the length of aC is 1. We can then construct the line CD, which as mentioned earlier will have length x.

It becomes apparent aAB forms a triangle similar to aCD, as both form isosceles triangles which are corresponding in their ratios of lengths. We can see $\frac{AB}{aB} = \frac{CD}{aC} \Rightarrow \frac{1}{x-1} = \frac{x}{1}$. Using algebraic manipulation,

we can then solve for x:

$$\begin{aligned}
\frac{x}{1} &= \frac{1}{x-1} \\
(x-1)\frac{x}{1} &= (x-1)\frac{1}{x-1} \\
x(x-1) &= 1 \\
x^2 - x - 1 &= 0 \\
-x^2 + x + 1 &= 0 \\
x &= \frac{-(-1)^2 \pm \sqrt{1-4(-1)(1)}}{2(-1)} && \text{(by \textit{the} quadratic formula)} \\
x &= \frac{1+\sqrt{5}}{2} = \varphi
\end{aligned}$$

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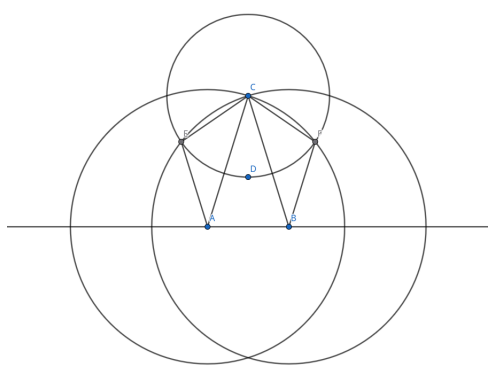
2. Given the result of the first prompt, describe how to construct a regular pentagon using only straightedge and compass. You may assume that φ has been constructed (that is, some length has been constructed that measures φ), and that 'numbers' (distances or lengths of line segments) can be transferred.

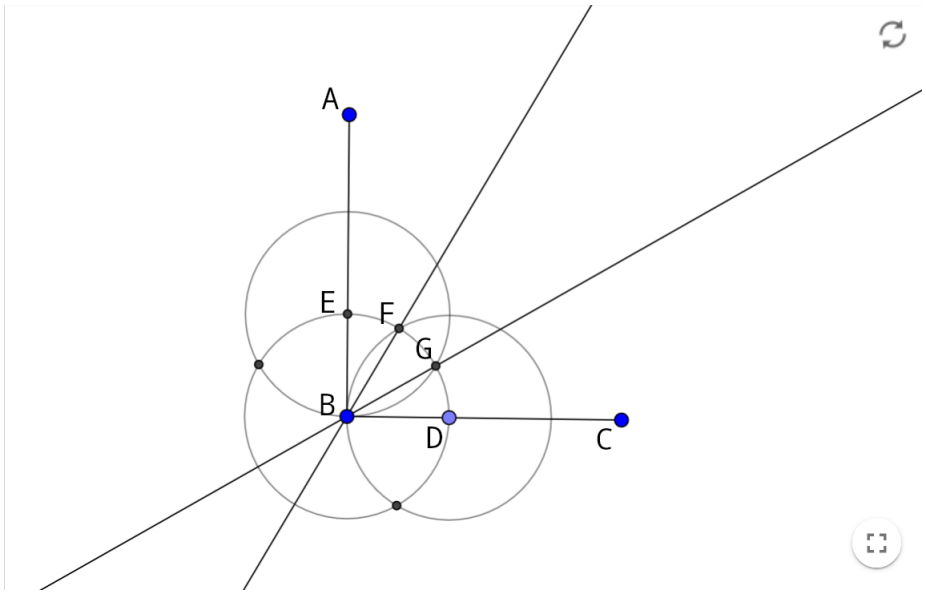
   Suggestion: Start by assuming I have the bottom side of the pentagon laid down and that its length is 1. Then use the Golden Ratio (which you may assume has been constructed and is transferable).

   Solution: Start with the bottom of the pentagon laid down with length 1 and denote its endpoints A and B. Using the compass, construct a circle of radius φ centered at B; repeat this construction centered at A. Call the circles centered at A and B c_1 and c_2, respectively. Denote the point of intersection of these circles C. Construct a circle centered at C with radius 1 denoted c_3. Then, find the intersection of c_3 and c_1, denote this intersection F, repeat for c_2 and c_3 denoting this intersection E. Construct the lines AB, AE, EC, CF and FB, which create a pentagon.



3. Trisect a right angle using straightedge and compass. A bullet-proof Euclid-style proof is not necessary, but the construction should be explained enough to be convincing.

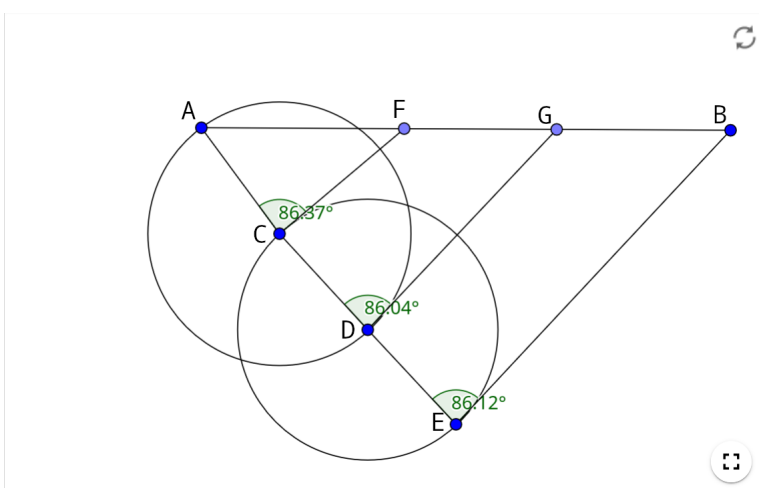
   Solution: Below is a construction I performed using Geogebra's *Geometric Constructions* tool, which allows for a digital equivalent of a straight edge and compass.



First I constructed the right angle by creating two tangent lines*. From there, mark point D, which will be our unit length, on line BC and create a circle which centers around point B, denote the intersection with the line AB as point E. Create another circle with radius of line BE centered at point E. Note the intersection of these circles in the right angle as point G. The points BEG form an equilateral triangle, meaning all interior angles of the triangle are 60 degrees. Recall the interior angle of a right triangle is 90 degrees. Since the triangle BEG shares the line BE with the right angle, we can see $\frac{BE}{BG} = \frac{2}{3}\frac{BE}{BD}$. If we mirror this construction by creating another circle centered at point D and then noting the intersection between this circle and our original circle as point F, then we can see we get an identical result where $\frac{BD}{BF} = \frac{2}{3}\frac{BD}{BE}$. Therefore, we can see $\frac{BE}{BF} = \frac{BF}{BG} = \frac{BG}{BD} = \frac{1}{3}\frac{BE}{BD}$, meaning the lines BF and BG trisect the angle created by lines BD and BE.

4. Trisect a line segment. For 6400 people: Provide a bullet-proof Euclid-style proof.

   Solution: Once again using Geogebra's *Geometric Constructions* tool, I created this figure using the straight-edge and compass

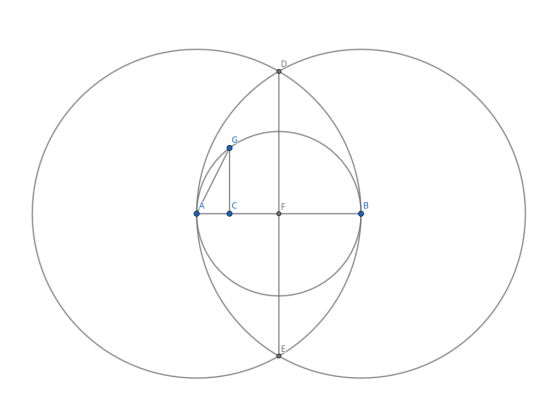


We label the endpoints of the line segment A and B. From there we draw a line segment out from A and denote the endpoint C. From here, using a compass we draw a circle whose radius is the length AC. We can then construct CD to be the line segment extending the line AC to the edge of the circle. From there

*I am assuming the right angle is given, so I used the angle tool to adjust the lines until the lines were tangent

we can draw a circle from C towards the point D, and construct the line and point DE and E respectively with the same technique as CD. From here, construct the line BE.

Using the angle of line BE, Construct lines between the points C and D which intersect with the line segment AB; we now have the lines CF and DG. We can then see we have triangles ABE, ACF, ADG which are all similar by construction. Thus, we can see $\frac{AC}{1} = \frac{AE}{3}$ and also $\frac{AD}{1} = \frac{2AE}{3}$. Recall the distance between points A, C, D, E is equal by construction. Since all the triangles ABE, ACF, ADG are similar, then we can see $\frac{AF}{1} = \frac{AB}{3}$ and $\frac{AG}{1} = \frac{2AB}{3}$. Therefore, the line segment is trisected by points F and G.

5. Pretend you are given an arbitrary length (magnitude or *number*) ℓ; using straightedge and compass (anything else would be uncivilized) construct $\sqrt{\ell}$. Provide an argument for your construction; 6400ers: Euclid-style bullet-proof proof.

   Solution: Construct a line with length $\ell+1$. Denote the endpoints A and B, and denote the point at $A+1$ on the line AB C. From here, find the midpoint by constructing circles with radius $\ell+1$ centered at the points A and B. Denote the intersection of these circles on the line AB as the point F. Construct a circle centered at point F with radius $\frac{\ell+1}{2}$, denote the circle c_1. Construct a line perpendicular to AB from c_1 to point C, denote the point this line intersects the circle as point G. Construct a per line GA. This line measures $\sqrt{\ell}$.